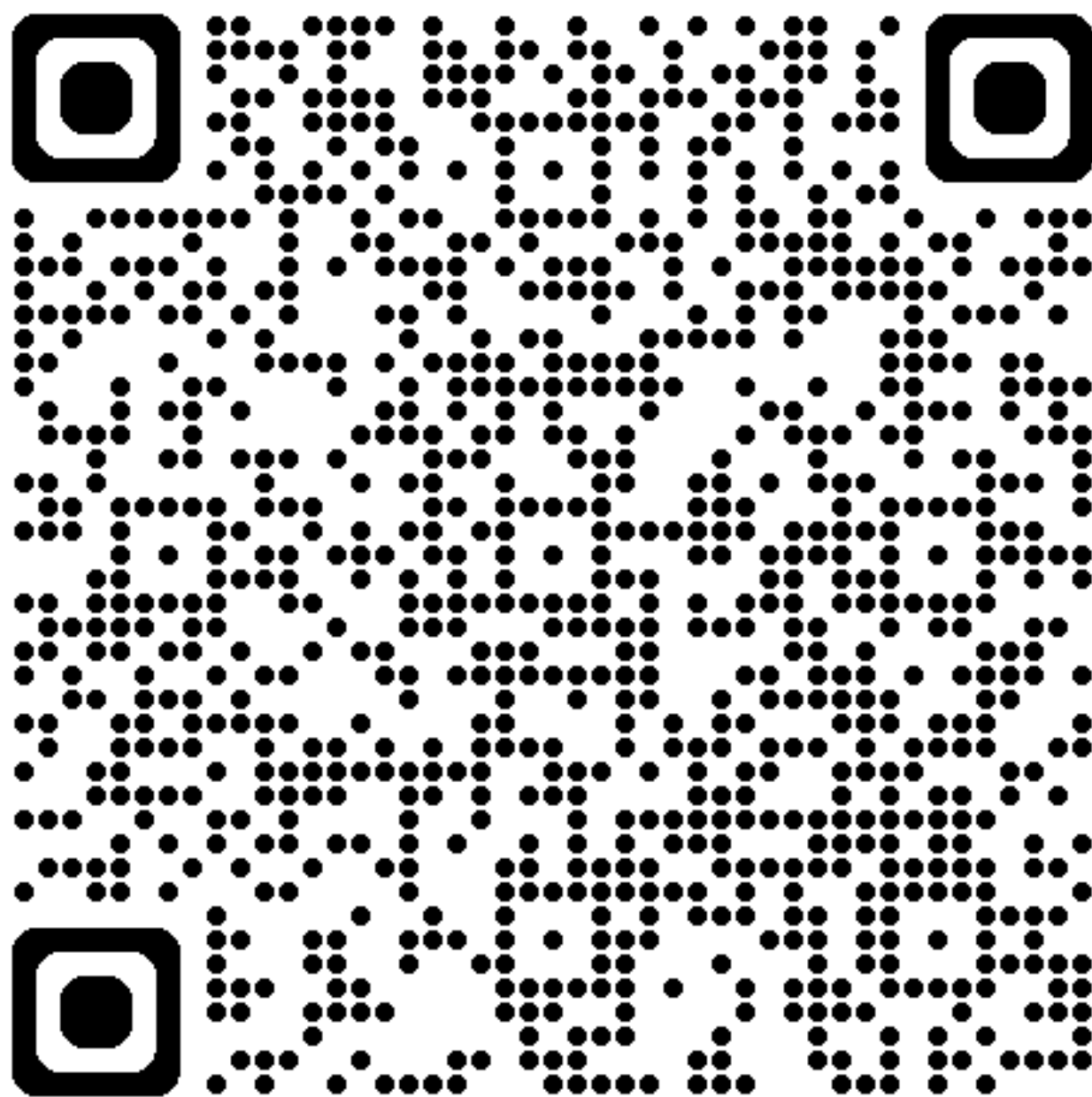




Put An Agent Inside Your App In 10 Minutes Or Less

With the GitHub Copilot SDK

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LEAN
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Outline

Foundation

- The GitHub Copilot SDK
- What's an agent?
- How the SDK works
- Demo – “10 minutes or less”

Application

- Why use the SDK?
- The cooler demo
- Taking it to production

What is the GitHub Copilot SDK?

- Open-source SDK to call GitHub Copilot (or other AI) from code
- BYOK support
- Official: Typescript, Python, Go, C#, Rust, Java
- Unofficial: Clojure, C++

What is the GitHub Copilot SDK?

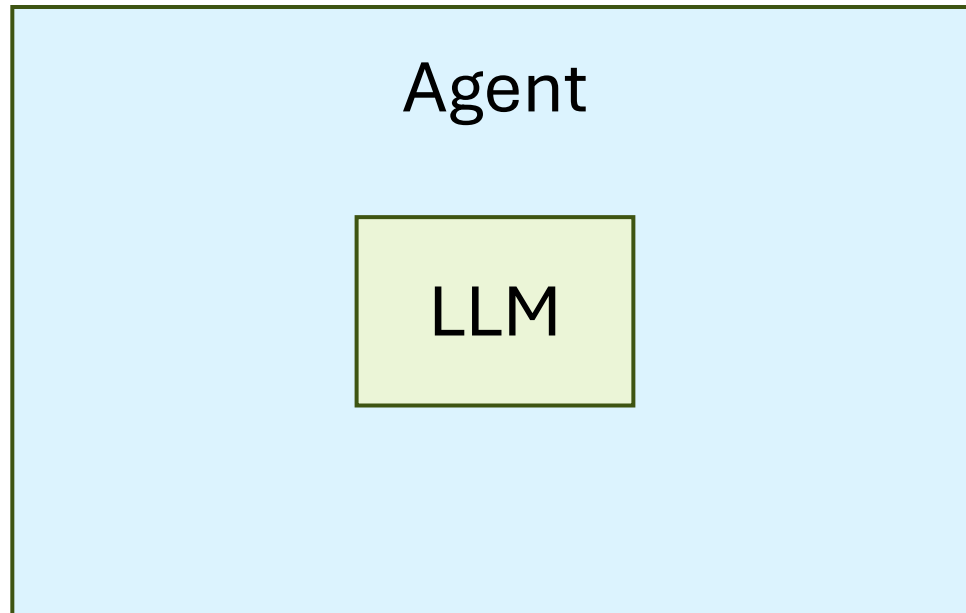
```
var client = new CopilotClient();  
await client.StartAsync();  
  
var session = await client.CreateSessionAsync(new SessionConfig  
{  
    Model = "claude-sonnet-4.6",  
    OnPermissionRequest = PermissionHandler.ApproveAll  
});  
  
var response = await session.SendAndWaitAsync(new MessageOptions  
{  
    Prompt = "What's the weather in San Antonio?",  
});
```

Wait!

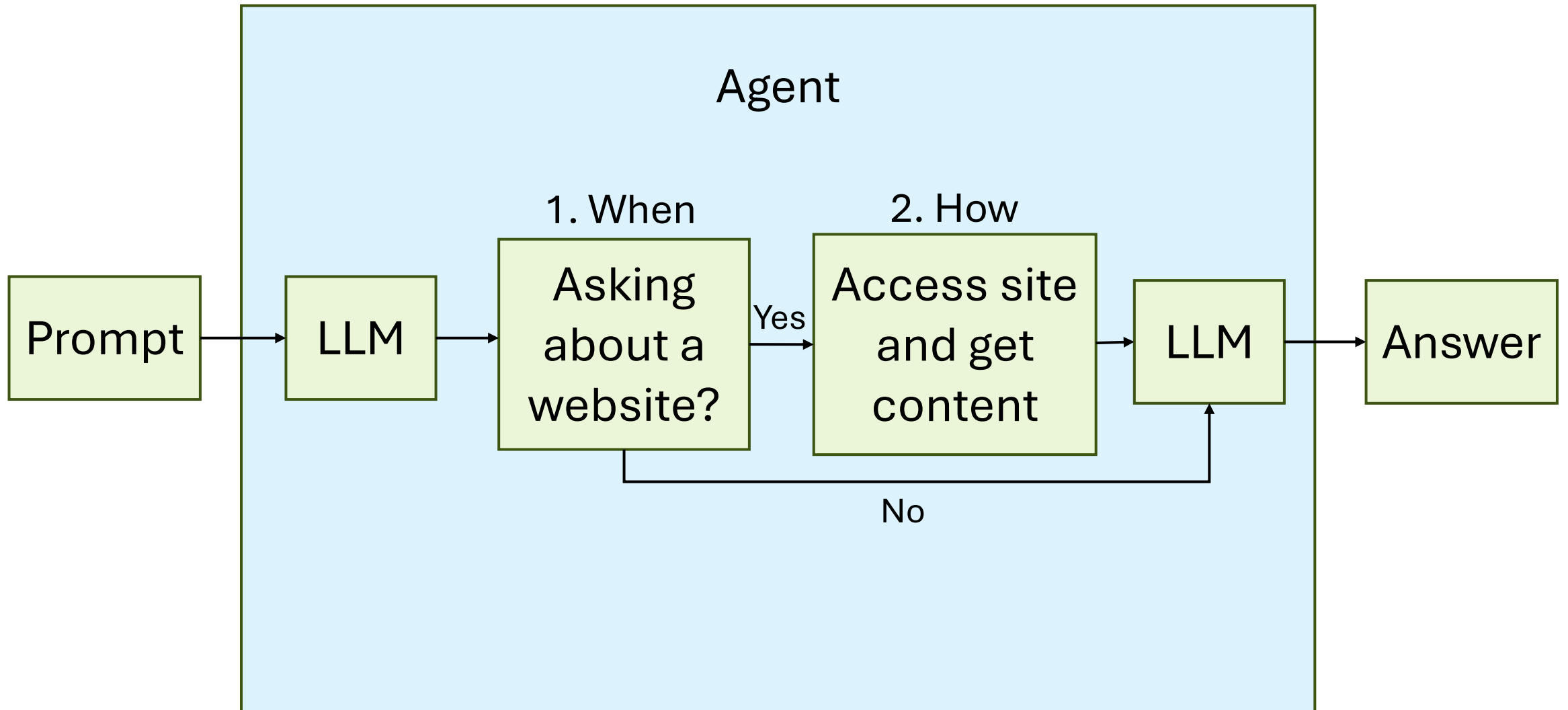
- What's an agent?
- Why do I want an agent in my app?



What's an agent?



What's an agent?



```
{
  "name": "web_fetch",
  "description": "Fetches a URL from the internet and returns the page as either markdown or raw HTML. Use this to safely retrieve up-to-date information from HTML web pages.",
  "parameters": {
    "url": {
      "type": "string",
      "description": "The URL to fetch"
    },
    "raw": {
      "type": "boolean",
      "description": "If true, returns raw HTML. If false, converts to simplified markdown (default: false)"
    },
    "max_length": {
      "type": "number",
      "description": "Maximum number of characters to return (default: 5000, maximum: 20000)"
    },
    "start_index": {
      "type": "number",
      "description": "Start index for pagination. Use this to continue reading if content was truncated (default: 0)"
    }
  },
  "required": ["url"]
}
```

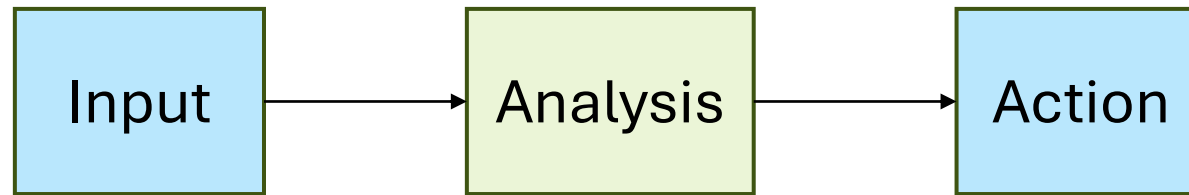
What's an agent?



What's an agent?

- LLM that we've given decision-making capabilities
- Through “tools” (just code)
- We tell the LLM:
 - When to use the tools
 - How to use the tools
- We let it iterate
- Agent = LLM + tools + loop

Why would I want an agent in my app?



1. Fix production errors – collect logs, clone source code, suggest a change, open a PR
2. Route support tickets – look at ticket info, judge severity & area, route to proper team
3. Daily brief – use info from tickets, repos, email, chats, & calendar, call out most pressing items, send email summary

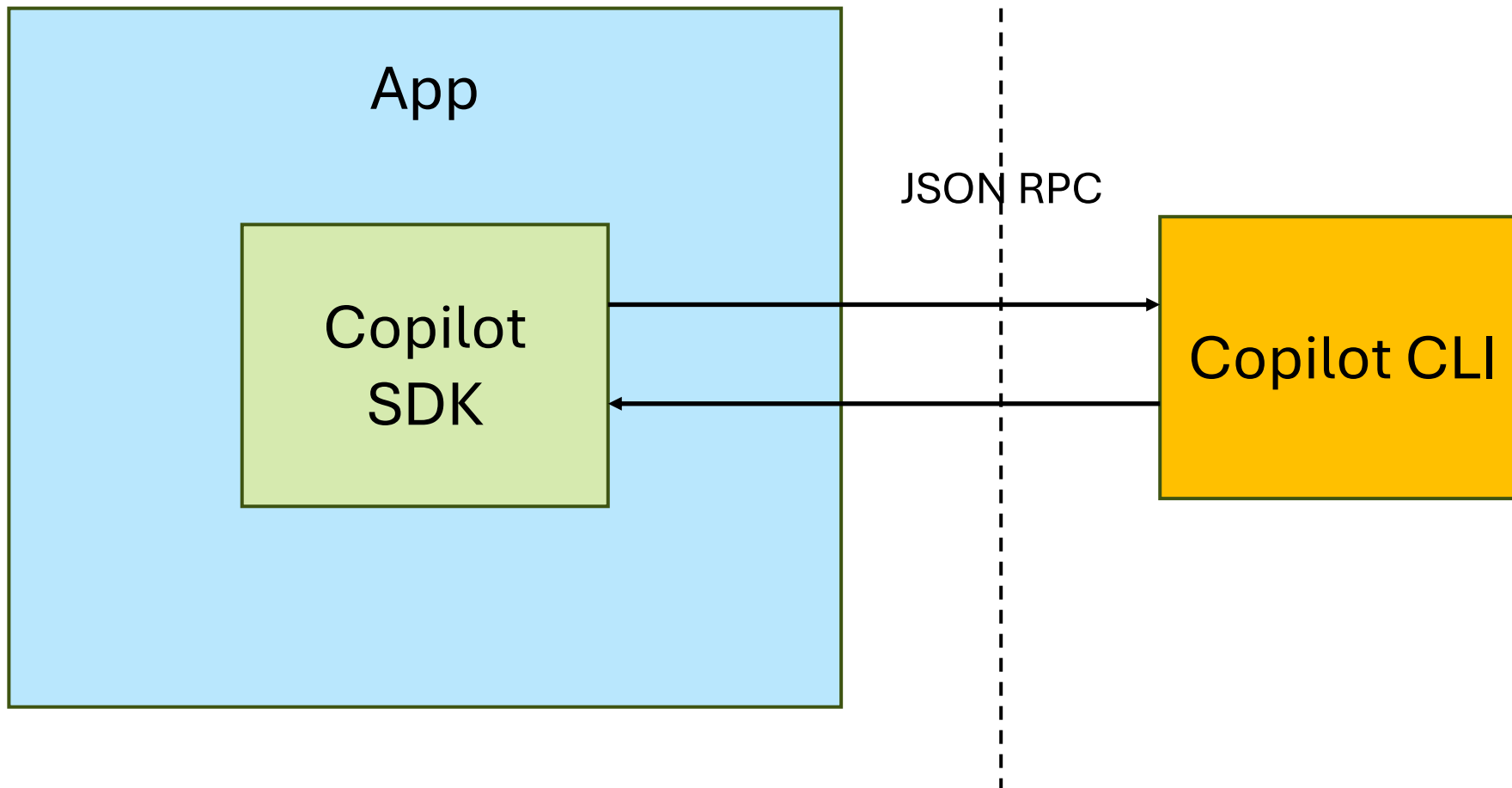
Coding demo

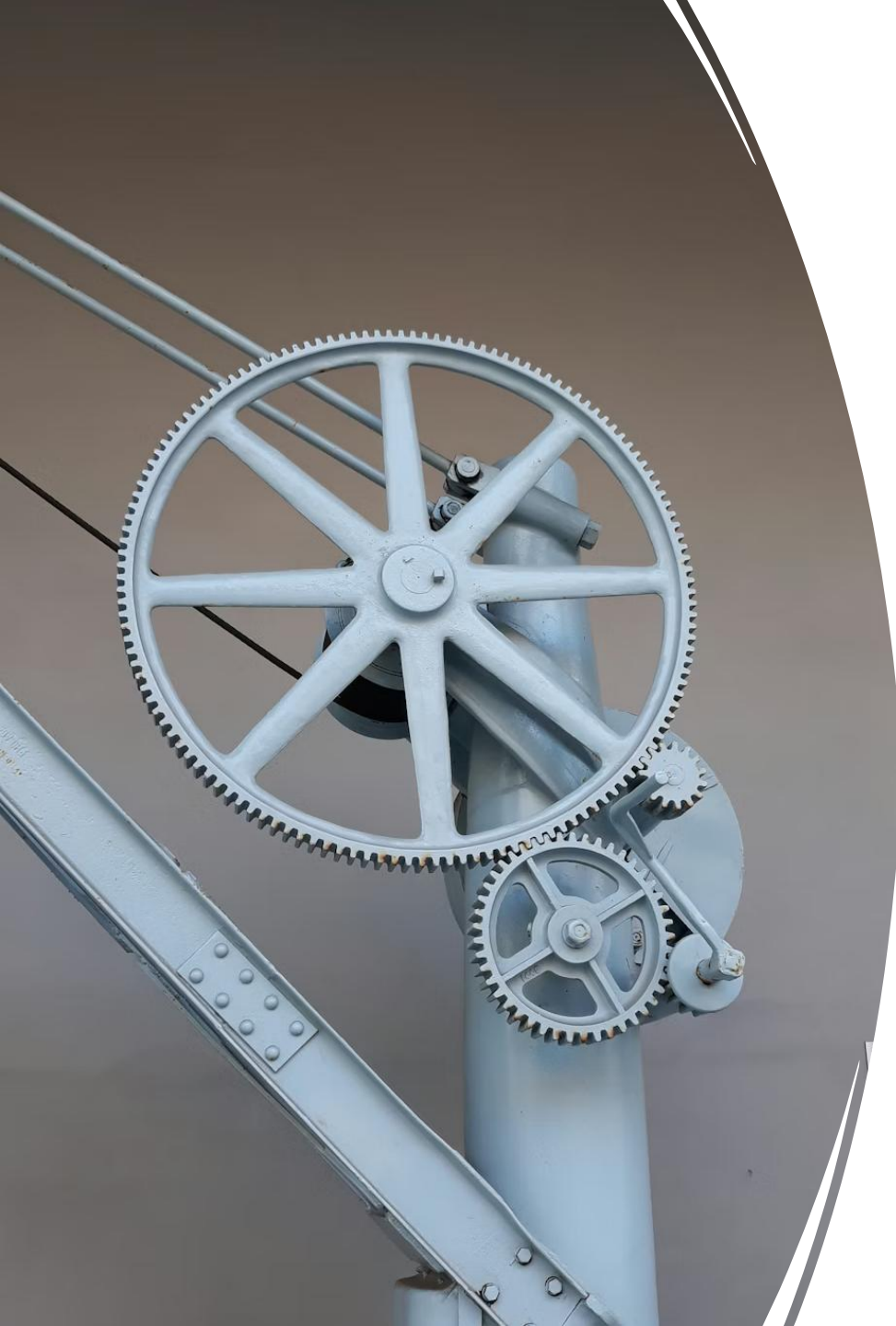
- Requirements
 - CLI installed
 - Copilot license/other API key

The SDK just uses the CLI

- SDK talks to the CLI through JSON-RPC

The SDK just uses the CLI





GitHub Copilot SDK features – CLI options

- Bundle CLI with app (default for Node, Python, C#)
- Run CLI installed on the host machine (default for Go, Java, Rust)
- Run on remote server



GitHub Copilot SDK features – CLI Parity

- Everything Copilot normally has in the CLI – agents, skills, MCP, etc.
- Small set of features not available
 - Protocol doesn't support it
 - Doesn't make sense for the SDK



GitHub Copilot SDK features - Auth

- Signed in CLI user (default)
- GitHub OAuth
- Env vars
- BYOK – use API key from OpenAI, Azure, Anthropic, etc.
 - Can also use local models
 - Don't need to be logged in

[illegible]

- [illegible]

```
[DisplayName("get_weather")]
```

```
[Description("Gets the current weather information in the requested city")]
```

1 reference

```
WeatherData GetWeather([Description("The name of the city to get the weather for")] string cityName)  
{  
    return new WeatherData(Temperature: 100, Condition: "It's too dang hot for this time of year");  
}
```

2 references

```
record WeatherData(int Temperature, string Condition);
```

```
var session = await client.CreateSessionAsync(new SessionConfig  
{  
    Model = "gpt-5-mini",  
    Tools = [AIFunctionFactory.Create(GetWeather)]  
});
```

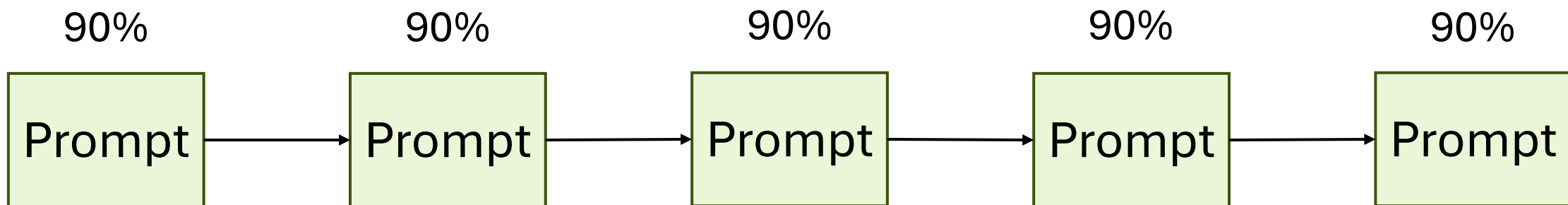


Wait! Can't I just do that with prompts?

- Yes!
- But...
- Prompt workflow vs. code workflow

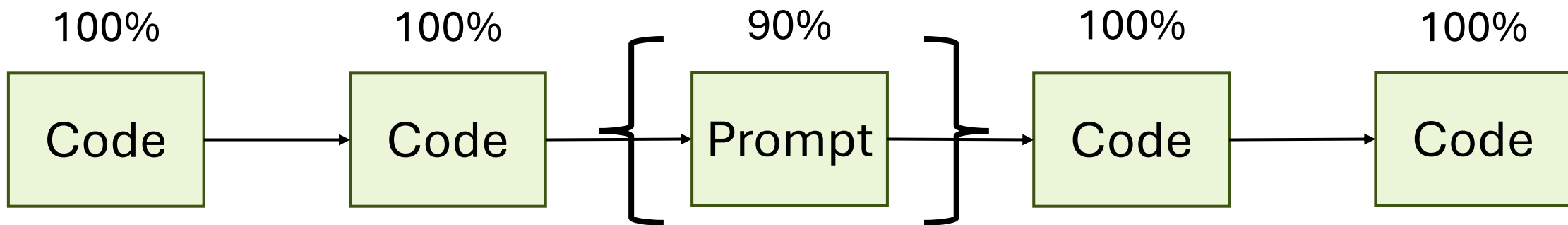
You can trust AI

But only a little



Quicker to set up

= 59% success rate

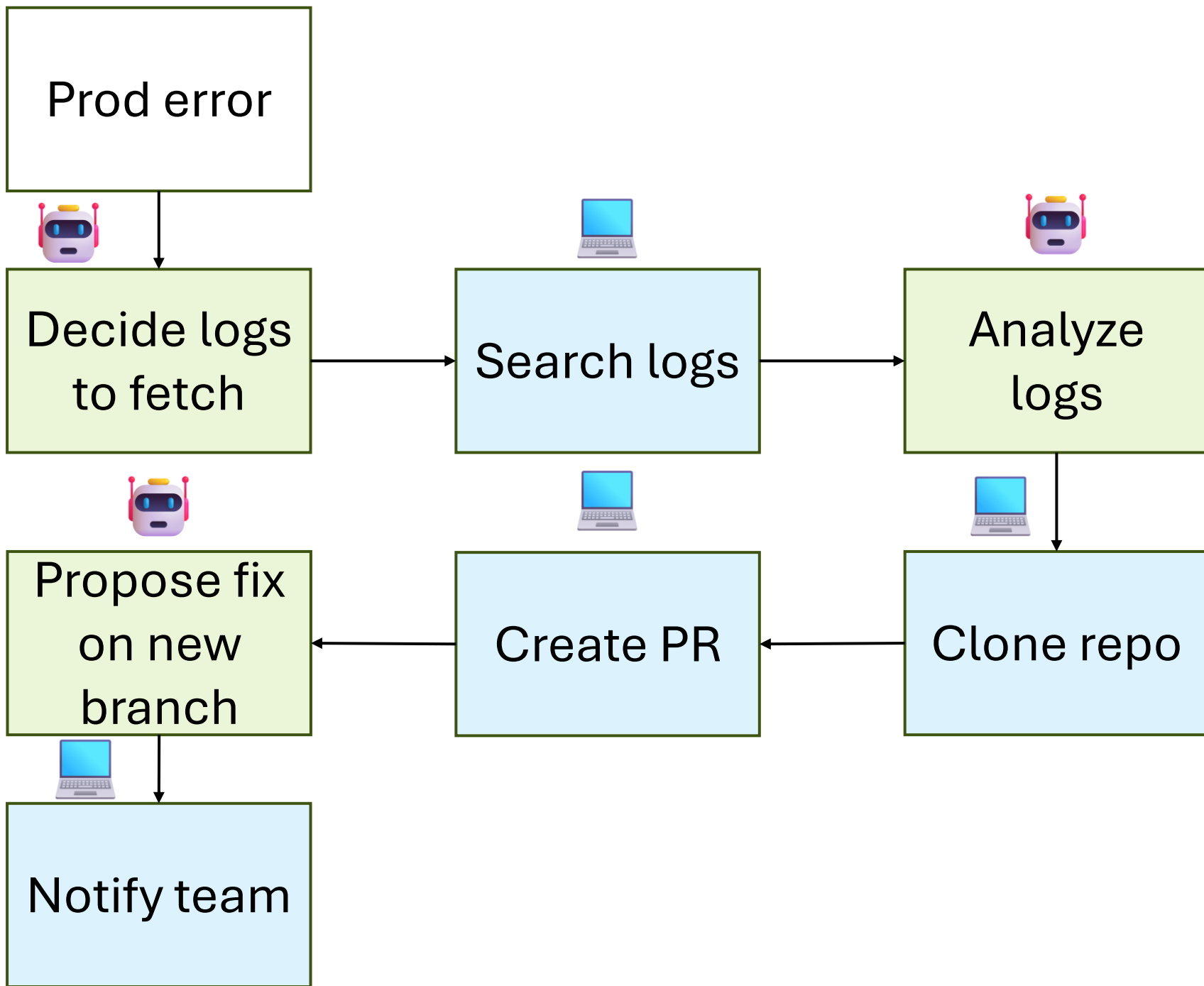


Cheaper overall

= 90% success rate

Some examples

- Pair programming/code review
- Review work before standup for a summary
- Brainstorming app ideas
- In-app recommendations



The cooler demo

```
mirror_mod = modifier_ob.  
#set mirror object to mirror  
mirror_mod.mirror_object  
operation == "MIRROR_X":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = False  
operation == "MIRROR_Y":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False  
operation == "MIRROR_Z":  
mirror_mod.use_x = False  
mirror_mod.use_y = False  
mirror_mod.use_z = True  
  
#selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier_ob.name))  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select  
  
print("please select exactly one mirror")  
  
--- OPERATOR CLASSES ---  
  
bpy.types.Operator):  
    "X mirror to the selected  
    object.mirror_mirror_x"  
    "Mirror X"  
  
def execute(self, context):  
    if context.active_object is not None:
```

Taking it to production

- Sandboxing
- You control the tools (no YOLO mode!)
- UX – streaming responses, wait times, etc.
- Log everything!

Authentication

- CLI user
- BYOK
- OAuth
- Env vars/tokens

CLI

- Bundled
- Local
- External

Users

- Single CLI
- Shared CLI
- CLI per user



Recap

- What's an agent?
- Why use an agent?
- GitHub Copilot SDK – CLI, features, auth
- Coding demo – “10 minutes or less”
- Why use the SDK over prompts?
- Coding demo – realistic example
- Prod considerations

Resources

- <https://github.com/github/copilot-sdk/>
- <https://github.com/github/awesome-copilot/tree/main/cookbook/copilot-sdk>

Thank you!

